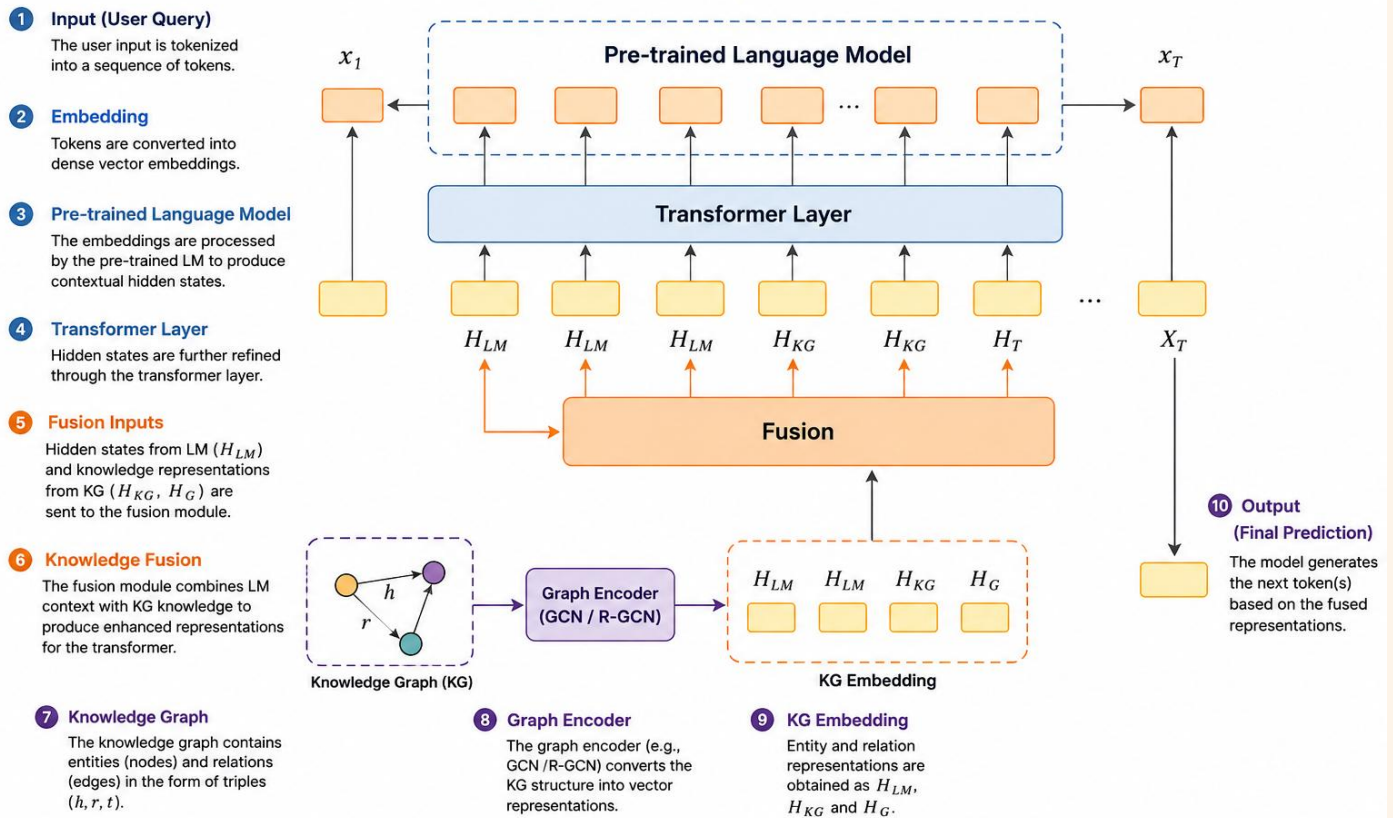


Working Flow of the Original Architecture



STEP-1: IDENTIFY THE PROBLEM

Researchers first analyzed the major problems faced by Large Language Models (LLMs).

These problems include:

- Hallucinations
- Weak structured reasoning
- Difficulty in multi-hop reasoning

STEP-2: PROPOSE A SOLUTION

Instead of relying only on the LLM's internal knowledge, the researchers proposed using an external source of structured knowledge called a **Knowledge Graph (KG)**.

The main idea was to combine:

- Context understanding from the LLM
- Structured knowledge from the Knowledge Graph

STEP-3: DESIGN THE ARCHITECTURE

The researchers designed an architecture containing two main components:

1. Pre-trained Language Model

- Understands the user's question.
- Learns the meaning and context of words.
- Generates contextual representations.

2. Knowledge Graph Guidance Module

- Takes information from the Knowledge Graph.
- Understands relationships between entities.
- Provides structured knowledge to the language model.

Both components work together instead of independently.

STEP-4: CONVERT THE KNOWLEDGE GRAPH INTO VECTORS

A computer cannot directly understand a graph.

So, the researchers first converted the Knowledge Graph into numerical vectors (embeddings).

The process is:

Knowledge Graph

↓

Triples (Head, Relation, Tail)

↓

Graph Neural Network (GCN / R-GCN)

↓

Knowledge Embeddings

Embeddings helps to preserve the relationships between entities.

STEP-5: EXTRACT CONTEXT FROM THE LANGUAGE MODEL

The user's input is passed through the Pre-trained Language Model. The language model generates contextual embeddings that represent the meaning of the sentence.

So now there are **two types of information**:

- Context embeddings from the LLM
- Knowledge embeddings from the KG

STEP-6: FUSE BOTH TYPES OF KNOWLEDGE

The researchers did not use the two outputs separately. Instead, they combined them using a **Fusion Module**.

The Fusion Module merges:

- Contextual knowledge (from LLM)
- Structured knowledge (from KG)

STEP-7: USE A GATING MECHANISM

Not every question requires the same amount of graph knowledge.

Therefore, the researchers introduced a **Gating Mechanism**.

Its job is to decide:

- How much information should come from the LLM.
- How much information should come from the Knowledge Graph.

The gate automatically adjusts the importance of each source according to the input question.

STEP-8: PASS THE FUSED INFORMATION TO THE TRANSFORMER

The fused representations are then sent to the Transformer layers.

The Transformer processes this enhanced information just like a normal language model but now has access to structured knowledge.

This helps it perform better reasoning.

STEP-9: FINE-TUNE THE ENTIRE MODEL

The complete architecture is trained on reasoning datasets.

During training:

- The Language Model learns better contextual understanding.
- The Knowledge Graph module learns better graph representations.
- The Fusion module learns how to combine both effectively.
- The Gating mechanism learns when to use graph knowledge.

STEP-10: GENERATE THE FINAL ANSWER

Finally, the model generates answer.